Some scientists think that this approach is archaic and needs to be rethought given what we know about the human mind and machinery now. Spearheaded by the Massachusetts Institute of Technology, some of the pioneers of the AI field have teamed up with new generations of thinkers to perform a massive "do-over" of the whole idea.

This time, they're determined to get it right – and, with the advantages of hindsight, experience, the rapid growth of new technologies and insights from the new field of computational neuroscience, they think they have a good shot at it.

The new project is called the "Mind Machine Project" or "MMP", a loosely bound collaboration of about two dozen professors, researchers, students and postdocs. It's led by Newton Howard, who came to MIT to head this project from a background in Government and Industry Computer Research and Cognitive Science.

According to Neil Gershenfeld, one of the leaders of MMP and director of MIT's Center for Bits and Atoms, one of the project's goals is to create intelligent machines – "whatever that means."

The project is "revisiting fundamental assumptions" in all of the areas encompassed by the field of AI, including the nature of the mind and of memory, and how intelligence can be manifested in physical form, says Gershenfeld, professor of media arts and sciences. "Essentially, we want to rewind to 30 years ago and revisit some ideas that had gotten frozen," he says, adding that the new group hopes to correct "fundamental mistakes" made in AI research over the years.

The birth of AI as a concept and a field of study is generally dated to a conference in the summer of 1956, where the idea took off with projections of swift success. One of that meeting's participants, Simon, predicted in the 1960s, "Machines will be capable, within 20 years, of doing any work a man can do." Yet two decades beyond that horizon, that goal now seems to many to be as elusive as ever. It is widely accepted that AI has failed to realise many of those lofty early promises.

Fixing what's broken

Gershenfeld says he and his fellow MMP members "want to go back and fix what's broken in the foundations of information technology." He says that there are three specific areas – having to do with the mind, memory, and the body – where AI research has become stuck, and each of these will be addressed in specific ways by the new project.